Midterm Practice Exam 2

Same format as the real midterm: four parts, 100 points, 100 minutes. Calculator and one A4 sheet of notes allowed. Show your working. Short answers: one to three sentences.

Scenario for Parts B to D. A gym chain wants to contact members who are likely to cancel their membership within 60 days and offer them a free month. The offer is the action. A missed canceller (a member who would have stayed with the offer but was not contacted, and left) costs **4,000**. An unnecessary offer (to a member who would have stayed anyway) costs **500**. All money is in fictional units.

Formula sheet (as printed with the exam): Accuracy (TP+TN)/N · Sensitivity TP/(TP+FN) · Specificity TN/(TN+FP) · Precision TP/(TP+FP) · Total cost C_FN×FN + C_FP×FP · Break-even p* = C_FP/(C_FP+C_FN) · z = b0 + b1·x1 + b2·x2, p = e^z/(1+e^z) · odds = p/(1−p), odds ratio = e^b · ŷ = b0 + b1·x1 + b2·x2. "Flagged" means the model recommends the offer.

Part A · Concept checks (12 points: items 1–8 multiple choice, 1 point each; items 9–10 true/false with a one-sentence reason, 2 points each)

1. Adding the field *retention_offer_sent* (an offer staff send after a member phones to cancel) raises test AUC from 0.74 to 0.97. What should the chain do?

(A) Deploy: 0.97 is excellent.
(B) Lower the threshold to use the extra signal.
(C) Check when each input becomes known, and drop inputs that exist only after a member has started to cancel.
(D) Collect more rows so the effect is estimated more precisely.

2. The model has AUC 0.90 on members of existing branches. It will be used at three new branches in another city. What is the best evidence that it will work there?

(A) Good ranking and calibration on recent members of the new branches whom the model has never seen.
(B) AUC on a random 20% split of the existing file.
(C) Training AUC.
(D) A model with more inputs.

3. The team trained on a 50/50 resampled file. AUC is unchanged, but the average predicted probability is 0.45 while the real cancellation rate is 12%. What should they do before applying the break-even threshold?

(A) Nothing: AUC did not change.
(B) Use 0.5 as the threshold because the data are balanced.
(C) Remove the members who cancelled.
(D) Recalibrate the probabilities to the real base rate on untouched recent data, or set the threshold by rank.

4. *visits_per_month* and *classes_booked* correlate 0.93. In a model with both, *classes_booked* gets a negative coefficient. What is the best reading?

(A) Booking classes makes members cancel.
(B) The two inputs carry almost the same information, so their separate coefficients are unstable; predictions can still be fine.
(C) The model must be discarded.
(D) Drop visits_per_month because its coefficient is smaller.

5. Members on a student plan cancel more overall, but once contract length is in the model, the student coefficient turns negative. Why?

(A) A coding error.
(B) Students are always safer customers.
(C) Students are more often on short contracts, which drive cancellation; at the same contract length students cancel less.
(D) The sample is too small.

6. Lowering the threshold from 0.50 to 0.20 will usually:

(A) raise recall, lower precision and flag more members.
(B) raise both recall and precision.
(C) lower recall and raise precision.
(D) change nothing, because AUC stays the same.

7. Model X has test AUC 0.85 and Model Y 0.83. Which do you deploy?

(A) X, because higher AUC is always better.
(B) Y, because it is simpler.
(C) Whichever has more inputs.
(D) Compare them at the threshold you would actually use: cost per 1,000, capacity, calibration and how easy each is to explain.

8. An analyst tests 30 candidate variables at $\alpha = 0.05$ and reports the 2 that are significant. If none truly matters, roughly how likely is at least one false positive?

(A) 5%
(B) about 79%
(C) 30%
(D) 2%

9. True or false, with one sentence: if an input has $p < 0.001$, changing that input will change cancellations.

10. True or false, with one sentence: doubling the number of rows fixes the problem that members who joined through the new app are missing from the training file.

Part B · Reading and comparing regression models (34 points)

A logistic regression predicts whether a member cancels within 60 days.

Term	Coefficient	Std. error	p-value
Intercept	−1.80	0.12	< 0.001
contract_monthly [Yes] (vs annual)	0.90	0.20	< 0.001
visits_per_month (per visit)	−0.30	0.05	< 0.001

1. (3 p) Compute the predicted probability of cancelling for a member on a monthly contract who visits 2 times per month. Show your working.

2. (5 p) (a) Interpret the coefficient −0.30 in terms of odds, and state what is held fixed. (b) A colleague says: "contract_monthly (0.90) matters three times more than visits (−0.30)." Explain why that comparison is not fair.

3. (6 p) A manager says: "visits_per_month has p < 0.001, so sending reminder emails that make members visit more will cut cancellations." (a) Is this supported? Explain. (b) Describe an A/B test that could support it: who is randomly assigned, what is compared, and why random assignment matters. (c) Name one thing you would check before trusting the result.

4. (4 p) The training data contain only members who joined in person at a branch. The chain has just launched app-only sign-ups. What does this mean for scoring app members, and what would you do?

5. (4 p) A linear regression fitted on members with 0 to 20 visits per month gives: monthly spend on extras = 12 + 1.5 × visits_per_month, with training $R^2 = 0.30$ and residual standard error 8. (a) Interpret 1.5 in words, with units. (b) Use the residual table below: what do you conclude about the straight line and about precision for heavy users? (c) Predict the spend of a member with 30 visits per month and say whether you trust it.

6. (4 p) (a) In Model 3 (table below) neither coefficient is significant, although each was significant alone. Why? (b) Can you say that one more booked class adds 0.75 to spend with visits held fixed? (c) What would you do if the goal is prediction, and what if it is explanation?

7. (4 p) (a) Which model overfits, and how can you tell? (b) Which model do you recommend, and why? (c) Model 2's extra inputs come from a monthly satisfaction survey that only 55% of members answer. Name one thing you would check before relying on it.

8. (4 p) (a) Give an approximate 95% confidence interval for contract_monthly (coefficient ± 2 standard errors), convert it to odds ratios, and say what it means. (b) What does the interval tell you that the p-value does not? (c) The model is refitted on 2 million rows. The visits coefficient is now −0.02 with p < 0.000001. Is visits now an important predictor? Explain.

Table for item 5. Residuals of the spend line (actual minus predicted), by visit band.

Visits per month	Members	Mean residual	Spread (SD)
0–4	150	+2.1	3.0
5–9	300	−0.8	5.0
10–14	200	−1.4	7.5
15–20	50	+2.6	11.0

Table for item 6. Linear regressions for monthly spend. visits_per_month and classes_booked correlate 0.93.

Model	Inputs	Coefficients (standard error)	R^2
1	visits	visits 1.50 (0.20)	0.30
2	classes_booked	classes 2.00 (0.30)	0.29
3	visits and classes_booked	visits 0.95 (0.55); classes 0.75 (0.80)	0.31

Table for item 7.

Model	Inputs	Training AUC	Test AUC
1	contract, visits	0.74	0.73
2	Model 1 plus 5 survey measures	0.81	0.79
3	Model 2 plus 60 further variables	0.98	0.67

Part C · Cost and threshold (29 points)

A test set has 1,000 members. 120 of them cancelled. "Flag" means the model recommends the offer.

Threshold	Members flagged	Flagged and cancelled	Not flagged and cancelled	Flagged, did not cancel	Not flagged, did not cancel
0.50	50	36	84	14	866
0.20	180	78	42	102	778
0.10	350	102	18	248	632

1. (5 p) At threshold 0.20, compute sensitivity (recall) and precision, and say in one sentence what each means for the chain.

2. (7 p) (a) Compute the missing cost in the table below (4,000 per missed canceller, 500 per unnecessary offer). (b) Which policy would you choose, and why? (c) The model at 0.50 is 90.2% accurate and "offer nobody" is 88.0% accurate. What does that comparison hide? (d) The chain has 25,000 members a year at risk of being scored, and running the model costs 2,000,000 a year. Taking the model at 0.10 against the best baseline, is it worth running, and at what running cost would you stop?

3. (5 p) (a) Compute the break-even probability. (b) From 0.50 to 0.10, accuracy falls from 90.2% to 73.4%, yet cost falls. Sensitivity rises from 0.30 to 0.85 and specificity falls from 0.98 to 0.72. Explain why cost and accuracy move in different directions.

4. (7 p) The retention team can contact at most 200 members per 1,000. The CFO says a missed canceller may cost anywhere from 2,000 to 8,000. Use the second table below. (a) Which thresholds are feasible? (b) Recommend one threshold in two sentences and name the mistake you can afford. (c) At a missed-canceller cost of 8,000, how much would the chain gain per 1,000 members if it could contact 350? And at 2,000? Is extra capacity worth asking for?

5. (5 p) The figure shows the model's ROC curve with three points P, Q and R, one for each threshold. (a) Match 0.50, 0.20 and 0.10 to P, Q and R. (b) What does the dashed diagonal show? The AUC is 0.81: say in one sentence what that means. (c) The break-even probability is about 0.11. Which point do you choose, and why can the ROC curve alone not tell you the best point?

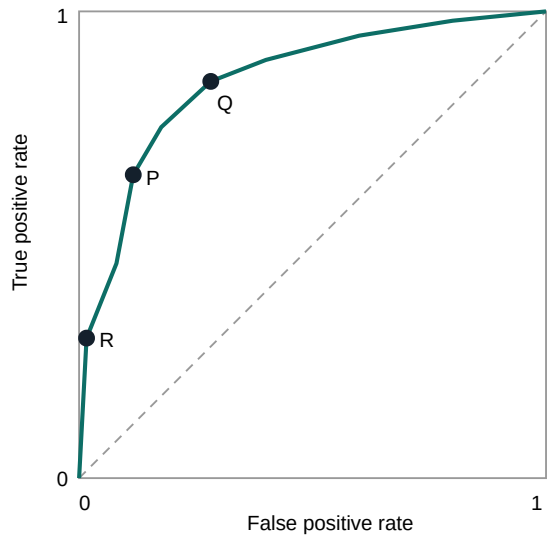
Table for item 2. Total cost per 1,000 members.

Policy	Total cost
Offer nobody	480,000
Offer everybody	440,000
Model at 0.50	343,000
Model at 0.20	219,000
Model at 0.10	(to compute)

Table for item 4. Total cost per 1,000 members, by what a missed canceller really costs.

Threshold	Members flagged	Cost at 2,000	Cost at 8,000
0.50	50	175,000	679,000
0.20	180	135,000	387,000
0.10	350	160,000	268,000

Figure. ROC curve (x = false positive rate, y = true positive rate).



Part D · Audit the AI and decide (25 points)

An AI assistant wrote a summary of the retention project for the board. You are the reviewer.

Item 1 (10 points, 2 each). For each claim, write Supported or Not supported, and give one reason. Use Parts B and C as evidence.

- Claim 1: "The spend regression has $R^2 = 0.58$ on the training data, so it will explain 58% of next year's spend."
- Claim 2: "Members who booked a personal-trainer session cancelled far less, so offering free sessions will cut cancellations."
- Claim 3: "The model is 89% accurate on the test set, so it is ready to use." (The test cancellation rate is 12%.)
- Claim 4: "At threshold 0.20, the model flags 78 of the 120 members who later cancelled."
- Claim 5: "The AUC is 0.81, so 81% of the members we flag will cancel."

Item 2 (7 points). At threshold 0.20, the notebook prints expected and actual cost for two groups.

Group	Members	Expected cost	Actual cost
Branches opened in 2026	250	45,000	110,000
Existing branches	750	150,000	160,000

(a) Which group do you trust less, and why? (b) Name one check you would run. (c) What result would make you change the threshold, and in which direction?

Item 3 (8 points). Write the JUDGMENT and ACTION blocks of a board memo, at most 40 words each. They must agree with your numbers and claim no more than the evidence shows. The ACTION must name one number you will monitor after launch.